

Problem 1

Compute $\sqrt[4]{i}$ and $\sqrt[4]{-i}$

Proof.

□

Problem 2

Solve the quadratic equation

$$z^2 + (\alpha + i\beta)z + \gamma + i\delta = 0$$

Proof.

□

Problem 3

Show that the system of all matrices

$$\begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$$

with the standard matrix addition and scalar multiplication is isomorphic to $\mathbb{C}$

Proof.

□

Problem 4

Prove that

$$\left| \frac{a-b}{1-\bar{a}b} \right| = 1$$

if either $|a|=1$ or $|b|=1$

Proof.

□

Problem 5

Prove that

$$\left| \frac{a-b}{1-\bar{a}b} \right| < 1$$

if $|a|<1$ and $|b|<1$

Proof.

□

Problem 6

If $|a_i| < 1$, $\lambda_i \geq 0$, $i = 1, \dots, n$ and $\lambda_1 + \dots + \lambda_n = 1$

Show that

$$|\lambda_1 a_1 + \dots + \lambda_n a_n| < 1$$

Proof.

□